

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EE 322 — Past Questions & Materials
by Renance

Course: EE 322

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — EE 322

Level 300 • Second Semester • PastQuestion

SITY OF PETROLEUM RESOURCES, EFFURUN

E OF ENGINEERING AND TECHNOLOGY

ENT OF ELECTRICAL AND ELECTRONICS ENGI

COURSE CODE: EEE 322

NEERING

WMENATION: Second Semester

vm TITLE: Electrical Power Systems Principles

3) TIME ALLOWED: 3 Hours

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_ Figure 1: NORMINAL-z MEDIUM TRANSMISSION LINE

j). Using nominal- 2 medium transmission line shown in Figure 1, show that for

the generalized circuit constants, 10marks

$A=D=14+— B=Z,and $C$$

$=Y_t+ y_t$

te the units of the ee ee constants of a transmission line 4marks

n line delivering a load of

y IV. The transmission efficiency

f Given $A = D = 0.98 < 3$; $B=110< 75$ degrees ohm; $C = 0.0005 <$

y 80 degree mho ;

i b). What are the causes of voltage drop and line loss in a transmission

fi line. RATES i, at

QUESTION 3 La

3a. An overhead transmission line conductor having parabolic configuration |

weighs 1.925kg per meter length, area of cross section 2.2cm² and an

ultimate strength of 8000 kg/cm², when erected between supports 600m

apart and having 15m difference in height, determine the vertical sag from

the taller of the two supports which must be allowed so that the factor of ~

safety shall be 5 with the wire loaded due to 1kg of ice per meter and no

wind pressure. 9marks

3b. List at least five factors that affect sag in a transmission line 5marks

QUESTION 4 | ae

4a. A 400V, 50Hz, 3-phase line delivers 200kw at 0.8 pf lagging. It is desired to

raise the line power factor to unity by installing shunt capacitors. 7marks

Calculate the capacitance of each unit if they are connected in (i) star and (ii) delta

o* , 4b Deiiine the following: 7" 3marks

I) demand factor, I) load factor and III) diversity fetior

4c. What purposes are served by tariff? 5marks

QUESTION 5

5a. The Ugheli generating station has a maximum demand of 20MW, a load factor of 60 percent, plant capacity factor of 48 percent and a plant use factor of 80 percent. Find:

I) The daily energy consumed 3marks sar

II) The reserve capacity of the plant, 3marks

If) The maximum demand that could be produced daily, if the plants were running all the time and, 2marks P:wW

IV) The maximum energy that could be produced daily, if the plants were sk = running fully loaded. 2marks aw"

Sb. What are the significance of load factor and the diversity factor. 4marks

QUESTION 6

6a. Develop and explain a model that can represent electricity generation from various sources. 8marks

6b. What are the limitations of electricity generation by

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